

Article

Research on Grade Point Innovation and Grade Point Average Based on Deep Learning Networks and Evolutionary Algorithms for College Innovation Education

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Abstract: This study applies deep learning predictive networks and multi-objective decision-making algorithms to the context of innovation and entrepreneurship education, aiming to explore the characteristics of students in different majors regarding innovation and entrepreneurship. It also investigates how their inputs contribute to the enhancement of their innovation and entrepreneurship abilities, as well as the improvement of their academic performance. The researchers designed survey questions across four levels: internal and external factors, and subjective and objective factors. Longitudinal data are collected from 650 students at different grade levels. The results show a clear positive correlation between grade point innovation (GPI) and grade point average (GPA), and the relationship between students' learning characteristics and GPI and GPA is established using a deep network of deep kernel extreme learning machines. The strategies in the questionnaire are used as control variables to obtain learning strategies for different students using a multi-objective decision-making approach based on evolutionary algorithms. This study shows the effect of different resources on the improvement of students' innovation abilities and provides possible innovation strategy suggestions for different groups. The results of this study may contribute to the improvement of innovation and entrepreneurial curricula and educational methods.

Keywords: innovation and entrepreneurship; higher education; practical ability; undergraduate



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1. Introduction

Universities are the main force of innovation and entrepreneurship education. Meanwhile, students are the main body of participating in innovation and entrepreneurship education. The effectiveness of how universities implement innovation and entrepreneurship education has profound and far-reaching implications [1]. Following the West, China has shown positive support for the development of entrepreneurial universities, but is relatively cautious in actual action [2]. Although innovation and entrepreneurship education has been commonly carried out in universities, there are still many problems in its curricula, which are highlighted as the disconnect between curricula and entrepreneurship practice, the mismatch between innovation education and professional education, and the lack of formative evaluation of innovation and entrepreneurship [3]. These challenges not only lead to a waste of teaching resources but also become a bottleneck that restricts the enhancement of innovation and entrepreneurship education. How local universities can further build innovation and entrepreneurship programs that benefit all students has

become a new issue of the times. Relying on the recognition of school innovation and entrepreneurship points, this study tries to analyze the status quo of innovation and entrepreneurship education in local universities through the innovation and entrepreneurship courses, training projects, and competitions of university students in certain universities, and puts forward targeted suggestions for improvement and implementation cases.

1.1. Literature Review on Factors of Innovation

Innovation and entrepreneurship education is influenced by school policies, teachers' ability, students' motivation, etc. The promotion of innovation ability by these factors has always been a topic of interest.

- Students' innovation consciousness, innovation emotion, and innovation willingness are important parts of innovation and entrepreneurship education. Research results show that there is a significant positive correlation between innovation education and collaboration, between collaboration and strategic innovation, and between innovation education and strategic innovation [4]. In order to improve the innovation and entrepreneurial ability of college students and provide a new direction and platform for the innovation and entrepreneurship education model, it is necessary to integrate Internet thinking into the innovation and entrepreneurship courses of universities to guide, educate, and train college students. Student entrepreneurial teams can build long-term and sustainable competitive advantages by continuously accumulating and training of resources and capabilities. The entrepreneurial projects of college students' entrepreneurial teams are mostly high-tech industries, which belong to technology-intensive knowledge industries. High-tech industries need to cultivate innovation thinking and creativity through innovation education, rely on core assets and capabilities to achieve strategic innovation, grasp core values, build blue ocean markets, and accumulate more core resources. The research confirms the positive correlation between innovation education, teamwork, and strategic innovation, and provides a reference for the direction and model of innovation education in universities. To sum up, innovation and entrepreneurship education is a systematic project, which needs to comprehensively consider many aspects such as innovation consciousness, innovation emotion, innovation willingness, teamwork, strategic innovation, and Internet thinking. By organically combining these elements, we can provide students with a more comprehensive and systematic education on innovation and entrepreneurship, help them better adapt to market demand, and achieve the mutual development of individuals and society.
- Despite the growing recognition of the value of interdisciplinary learning in innovation and entrepreneurship education, there is still a paucity of relevant analytical literature. The research demonstrates the key role of individual interaction and behavior in shaping creativity and appeals to the academic community to pay attention to the microfactors in creativity cultivation rather than simply focusing on the external environment. The literature uses an interdisciplinary approach to examine the ways in which the University of Exeter leverages spaces for creativity and peer learning [5]. University administrators can use interdisciplinary education as a way to promote innovation in research. According to the literature [2], they can also integrate research into teaching by encouraging students to participate in practice-oriented research projects. Through strengthening cooperation between schools and enterprises, knowledge production and social application can promote each other. By integrating ideas, methods, resources and opportunities from the industry, we promote educational reform.

- Competition between disciplines plays an important role in cultivating innovation and entrepreneurial ability and is an important part of innovation and entrepreneurship education in universities. However, the role of competition between disciplines in innovation and entrepreneurship education needs to be further clarified. One study [6] incorporated competition-based learning into machine learning courses. By engaging students in innovation problem-solving challenges within information competitions, the study revealed that students' participation in online problem-solving competitions can improve their information technology abilities, while showcase competitions can enhance their competition ability. Another study [7] formed a curriculum model of "theory + experiment + practice" to guide students to participate in competitions. Innovation and entrepreneurship education should be coordinated with professional education. Professional education and high-quality humanistic education are added to the curriculum of innovation and entrepreneurship education, with the aim of cultivating students' innovation consciousness and lifelong learning habits. The competition project promotes the training mode for the sustainable development of education.
- To achieve systematic and comprehensive scientific evaluation, ref. [8] proposed an evaluation model based on swarm algorithms to optimize neural networks. The sustainability evaluation index system of innovation and entrepreneurship education for clean energy majors in colleges and universities is constructed from four aspects—environment, investment, process, and results—and the meaning of each evaluation index is explained. In [9], the authors aimed to identify the status quo of artificial intelligence in entrepreneurship education with a view to identifying potential research gaps, especially in the adoption of certain intelligent technologies and pedagogical designs applied in this domain. The collected research was analyzed from various perspectives, such as the definition of intelligent technology, research questions, educational purposes, research methods, sample size, research quality, and publications, with a focus on the adoption of AI in entrepreneurship education. The authors of [10] considered entrepreneurship and entrepreneurship career ability in a multiple-attribute decision-making model. A numerical model for evaluating the entrepreneurship and entrepreneurship career ability of local college students was constructed along with some decision comparisons to verify the method. A method for evaluating the performance of innovation and entrepreneurship education in Chinese technical universities is presented in [11], which is based on specific performance indicators, including patents filled, publications, awards in competitions, and acquired funding, and on certain non-specific ones, including organizational arrangements and satisfaction rates.
- In the context of engineering education professional certifications, the conceive–design–implement–operate (CDIO) and outcome-based education (OBE) approaches have gained increasing attention [12]. Engineering education puts forward new requirements for the teaching mode. Guided by the educational philosophy of CDIO and OBE, teaching reform has been implemented to better cultivate students' practical abilities, innovation abilities, and knowledge-integrated application abilities. The CDIO engineering education model with AI courses has been verified to be effective for students' learning [13]. The application of the CDIO model in educational contexts improved students' achievement of goals in AI courses. Students' teamwork ability, computational thinking ability, and critical and innovative thinking also improved. The flipped CDIO approach provides students with the opportunity to learn programs with convergent and divergent thinking to promote students' flow experience and sustainable learning [14]. Thus, cultivating students' innovative consciousness and development of good mental habits has become an important part of our country's higher

education research. Promoting the TRIZ-CDIO theory and innovation technology in engineering undergraduates' innovative ability training is of extremely important practical significance [15]. Although CDIO teaching reform practice has made some achievements in recent years, it is necessary to further explore and reform students' industrial practice ability and address the lack of supervision in the implementation of offline independent development projects [12].

The above-mentioned studies use different approaches to analyze innovation and entrepreneurship education, which is helpful to evaluation and improvement methods. In recent years, artificial intelligence (AI) methods have provided new avenues for influencing factors without education. AI methods are not only helpful for educational reform but are also highly significant to innovation and entrepreneurship education analysis. The authors of [16] explored the feature engineering applications of AI in education, highlighting its potential to enhance personalized learning and improve academic outcomes. Machine learning algorithms perform regression analysis of the data, with correlations in the datasets revealing complex connections between education, industry, job titles, training, and various other aspects. The importance of using data-driven strategies in policy development is highlighted, and further research on the use of AI methods in various educational settings is recommended to improve feature recognition and model performance. AI methods, such as deep learning neural networks and evolutionary algorithms, are less studied in innovation and entrepreneurial education. However, some experts and scholars have applied it in teaching research and reform. One study [17] proposes an effective OLSCA-KELM hybrid model to evaluate Chinese–foreign cooperative school programs. The main innovation of this method is to introduce an orthogonal learning mechanism into sine cosine algorithm, further balance global and local search capabilities, and establish the optimal prediction model of a nuclear learning machine to solve optimization problems. To further study the influence of mixed teaching strategies on ambidextrous innovation and evaluate its baseline effect, ref. [18] adopts the structural equation model (SEM) to study the influence of mixed teaching strategies on the innovation ability of students. In order to clarify the influencing mechanism of mixed teaching strategies, based on the ambidextrous innovation theory, students' innovation activities are divided into two types: exploitative innovation and exploratory innovation. The findings shed light on the potential impact of smart innovation and blended teaching strategies in educational contexts.

1.2. The Main Problems of Innovation Education

Based on the above literature analysis, this study summarizes the existing problems in innovation and entrepreneurship from the following four perspectives: construction of innovation curricula; analysis of innovation ability; evaluation of innovation achievement; and management of educational quality.

Construction of innovation curricula. In the classroom teaching of innovation and entrepreneurship education courses, more emphasis is placed on the transfer of theoretical knowledge, whereby teachers still play the role of the 'main body' and students are not more involved. There is no clear fit between the teaching objectives of innovation and entrepreneurship courses and the training objectives of talents in different majors. The course outline does not clearly point out the objectives and requirements of the innovation and entrepreneurship course system, and the detailed objectives of the course system's construction and even its reform are not clearly reflected. The objectives of innovation and entrepreneurship courses are vague and insufficiently refined, and thus cannot meet the needs of multiple common and individual goals.

Analysis of innovation ability. After receiving innovation and entrepreneurship education, the improvement effect of students' long-term innovation ability needs to be

further analyzed. College students are highly enthusiastic about participating in innovation and entrepreneurship competitions. However, most of them do not have a deep understanding of the competition, while a small number of students experience surprise competitions, and the quality of student parameter projects needs to be further improved. The support and coverage of college students' innovation and entrepreneurship projects are relatively high, but the students who apply for the projects are not fully prepared, and some students hope that their tutors will directly assign the projects, which leads to the phenomenon of cramming at the last minute. Research has shown that the overall level of student engagement in entrepreneurship courses in colleges and universities at this stage is not ideal, and the degree of active cooperative learning and curricular challenge only barely reaches the pass level.

Evaluation of innovation achievement. Innovation and entrepreneurship education is an important embodiment and concrete index of the realization of engineering education professional certification functions. The current evaluation index system is insufficient for meeting the developmental needs of colleges and universities [19]. Innovation and entrepreneurship education has not yet formed a specialised scientific and efficient evaluation system. The implementation standards of curriculum evaluation are consistent with the unified curriculum evaluation standards of schools. The key assessment of engineering education certification for innovation and entrepreneurship is universal and cannot be verified by the results achieved by top students in innovation and entrepreneurship competitions. Although colleges and universities supplement practical teaching with enterprise internships and participation in scientific research projects, they cannot achieve extensive assessment. Engineering education certification requires continuous student-centered improvement. Therefore, in the process of innovation and entrepreneurship education reform, emphasis is placed on the universal results of students' participation in innovation and entrepreneurship activities, with long-term tracking of students' innovation and entrepreneurship results at graduation after the completion of elective courses.

Management of educational quality. The quality control link of innovation and entrepreneurship education is weak, and an effective quality management model has not yet been formed. Traditional innovation and entrepreneurship education has not significantly increased students' innovation and entrepreneurial ability, nor has it produced significant innovation and entrepreneurship results. Courses, projects, and competitions in the process of innovation and entrepreneurship education have not been effectively integrated into higher education, especially in engineering education and teaching. In the process of professional engineering certification, the cultivation of students' innovation ability is emphasized, but a unified and effective standard has not been formed. The effectiveness of innovation and entrepreneurship courses and teaching activities lacks sufficient data support and empirical verification. Teachers use traditional teaching methods to emphasize the explanation of knowledge, ignoring students' learning behavior and effectiveness.

1.3. The Main Content of This Study

Figure 1 shows the main contributions of this paper. The correlation analysis of multiple factors is given in Section 2. Based on subjective and objective data, Pearson correlation analysis is used to determine the correlation between grades, credits, and students' investment in innovation and entrepreneurship. The correlation model between factors and decisions is given in Section 3. The deep kernel extreme learning machine (DKELM) is used to verify the correlation between students' innovation ability and each index. The model based on DKELM is used for multi-objective decision-making. Section 4 adopts the multi-objective optimization idea and evolutionary algorithm to evaluate the strategy of innovation and entrepreneurial ability. The multi-objective evolutionary algo-

rithm based on non-dominant sorting is used to solve the problem model to obtain more effective decision-making and improve students' innovation ability.

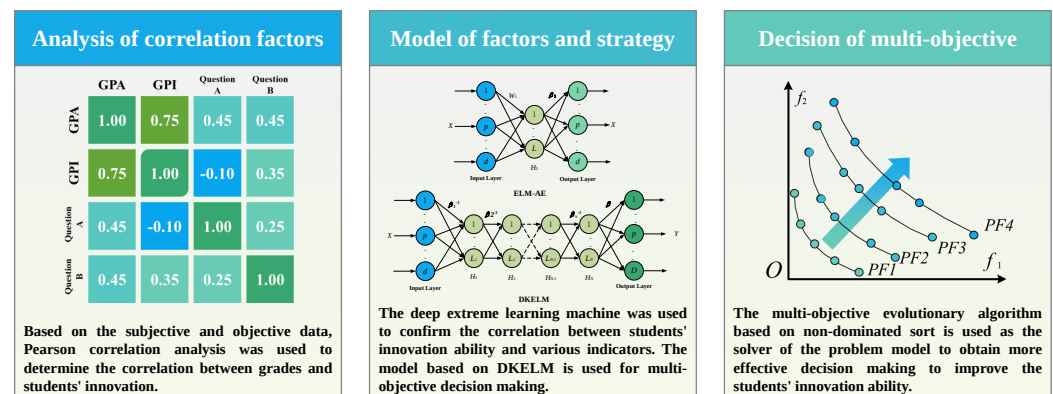


Figure 1. The main contributions of this work. Firstly, Pearson correlation analysis is used to determine grades and relationships among factors. Secondly, the DKELM method is used to construct a prediction model for innovation ability, learning ability, and evaluation factors and is used in multi-objective decision-making. Finally, the multi-objective optimization method is used to develop suitable strategies to improve students' innovation ability.

The main contributions of this work are summarized as follows:

- The correlation between innovation ability and influencing factors is analyzed to build a deep learning model.
- A deep learning model is established to reflect the relationship between the actions of the school and those of the student.
- An optimal strategy is obtained by evolutionary algorithm from the proposed deep learning model.

2. Correlation Factor Design and Analysis

In order to fully analyze the relationship between college students' innovation spirit, entrepreneurial consciousness, innovation entrepreneurial ability, and academic performance, this research designs questionnaires and question models on the basis of a large number of previous studies, drawing on and integrating the research and results of relevant scholars (Supplementary Materials).

2.1. Data Source

Since the year 2016, the school system has introduced grade point innovation ($GPI > 6$) as a prerequisite for students to qualify for graduation. It has set up the Innovation and Entrepreneurship Education and Guidance Center, which is fully responsible for the recognition and management of schools' innovation and entrepreneurship education. Universities have established an innovation and entrepreneurship management system to record all kinds of innovation activities and GPI obtained by each college student. They have achieved the unified management of the content of college students' innovation and entrepreneurship education, including disciplinary competitions, innovation and entrepreneurship training projects, papers, patents, social practice, and literary and sports quality training. This paper is mainly based on the grade point innovation of college students recorded in the system as a data source for the evaluation of college students' innovation and entrepreneurial ability. The minimum unit of the GPI is 0.5, and the value is an integer multiple of 0.5 point. The lower limit of the GPI is 6; there is no upper limit, and the upper limit recorded in the system is 90 points. Figure 2 shows the relationship between the GPI and the GPA of the students in the three grades of the four majors of

automation, electrical engineering and automation, intelligent science and technology, and measurement and instrument in our school, among which the freshmen are no longer considered because they have just entered the school. The automation and electrical engineering major and the automation major are traditional classical majors in our school in which more students achieve good results. The intelligent science and technology major is a newly created major with a small number of students enrolled and a relatively weak overall performance among them.

Taking study objects as an example, students with different majors have different performances in GPA and GPI. In general, GPA and GPI show a positive correlation trend; that is, students with better course scores also have a better performance in mass innovation ability. Most of the students achieved a $GPI > 6$ in the junior semester, verifying the innovation ability of all graduates.

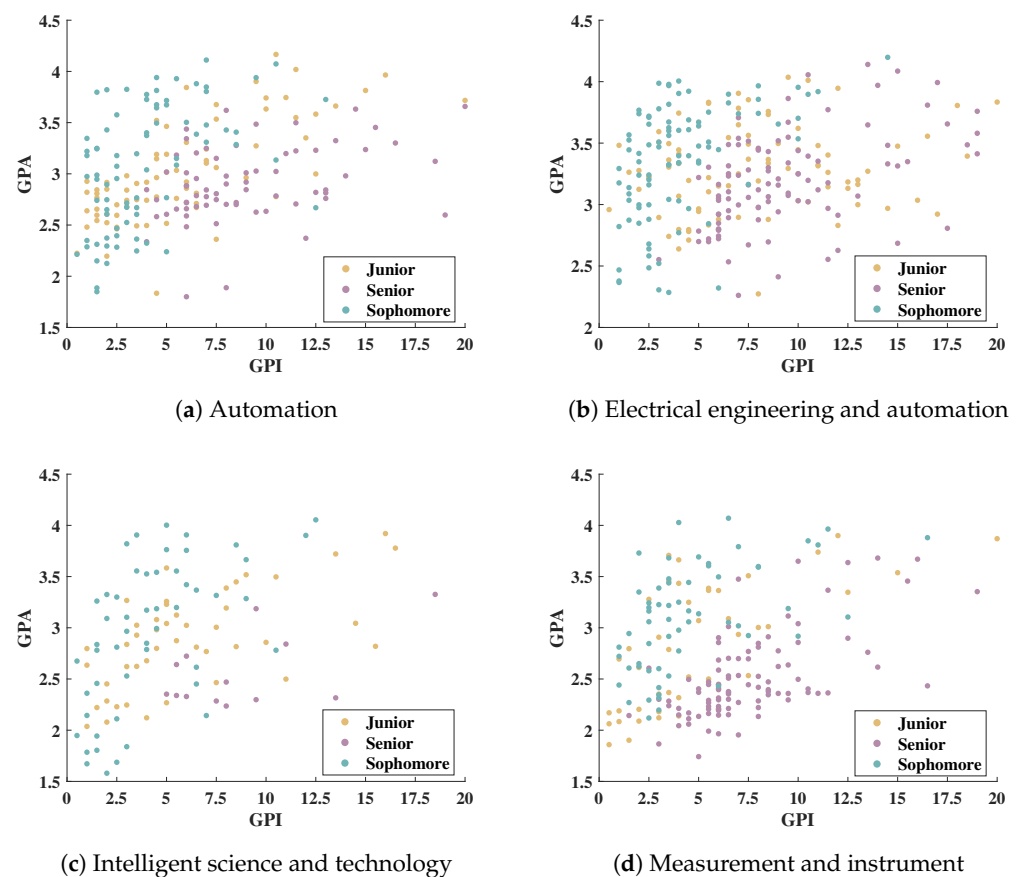


Figure 2. The relationship between GPA and GPI of students of different majors and grades.

2.2. The College Students Innovation Ability Model and Questionnaire Design

Based on the complexity of factors affecting college students' innovation and entrepreneurial abilities, there are significant differences in the evaluation index system for college students' innovation and entrepreneurial ability, especially among different types of universities. Therefore, when constructing the evaluation system, it is necessary to consider multiple dimensions to ensure that the evaluation can comprehensively and objectively reflect the innovation and entrepreneurial ability of college students. This paper focuses on two dimensions: subjective and objective factors, and internal and external factors. The subjective factors mainly involve the psychological and behavioral characteristics of college students, such as their personal interests and the proportion of time spent participating in innovation activities, while the objective factors include the influence of external condi-

tions such as innovation courses and innovation and entrepreneurship bases. The internal factors include college students' knowledge reserve, professional skills, teamwork ability, etc. The external factors involve external environmental factors such as the resources and policy system provided by the school. A total of 16 investigation and evaluation factors are designed. According to the evaluation index system for college students' innovation and entrepreneurial ability, a quantitative evaluation can be determined by referring to questionnaire, grade point innovation, and the type and proportion of achievements. Q1 to Q16 correspond to 16 evaluation factors, respectively, as shown in Figure 3.

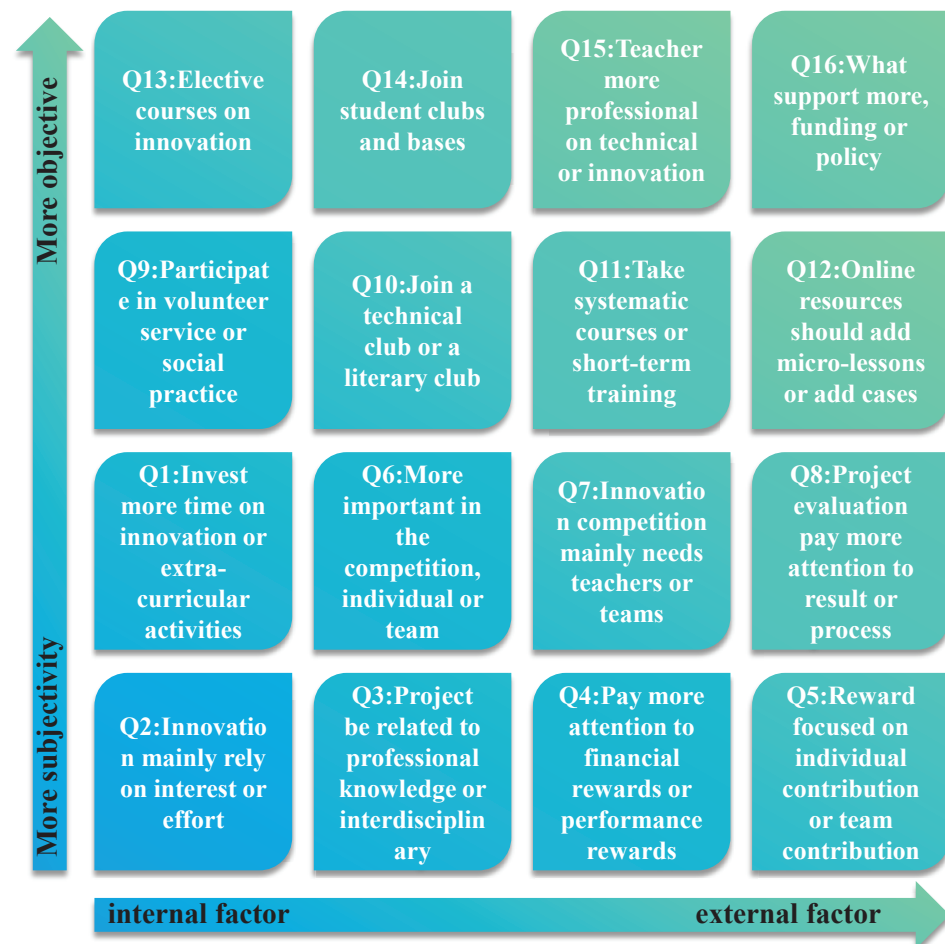


Figure 3. The design of questionnaire questions.

In order to produce effective questionnaires, the difficulty of the questions was simplified as far as possible in the process of the questionnaire's design. Using the idea of fuzzy rules, all problems were designed with five levels: negative big (NB), negative small (NS), zero (ZE), positive small (PS), and positive big (PB). On the basis of sending out 650 questionnaires, the valid questionnaires were screened according to the following principles: (1) remove the extreme values of plus or minus 5% of GPA and GPI; (2) choose the same answer for more than 50% of the questions. A total of 459 valid questionnaires were obtained. The relevant answer statistics are shown in Figure 4.

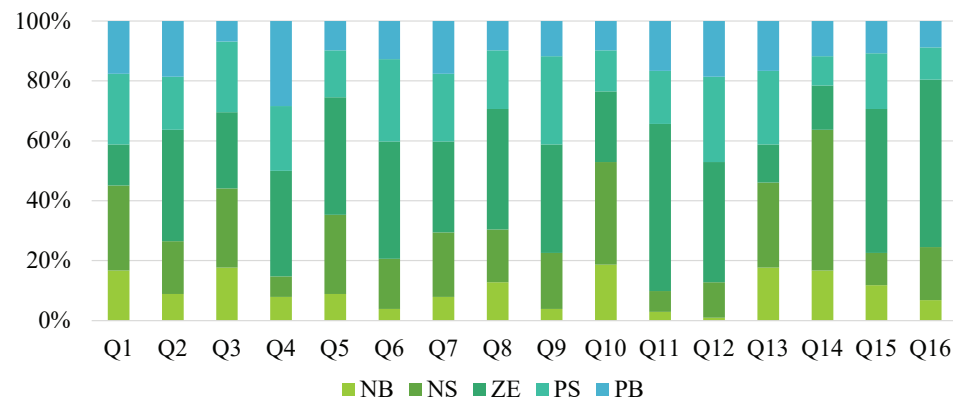


Figure 4. Questionnaire survey answer data statistics.

2.3. Correlation Analysis

Pearson correlation analysis is a tool used in statistics to measure the strength and direction of a linear relationship between two variables, the core purpose of which is to assess whether there is a positive, negative, or unrelated correlation between two variables, and to assess the extent of the correlation [20].

When Pearson correlation analysis is used to explore the relationship between college students' GPA, GPI, and 16 influencing factors, the main concern is whether these two variables increase or decrease synchronously when changing. For example, if the correlation coefficient is close to 1, then it can be considered that there is a positive correlation between the GPA and GPI; that is, students with a high GPA tend to obtain a higher innovation ability. This may indicate that students with a high GPA have excellent study performance and strong time-management ability, learning methods, or resource allocation ability, which may further aid them in achieving better results in extracurricular innovation and entrepreneurship activities. From another perspective, this positive correlation may also indicate that there is a mutually reinforcing mechanism between learning ability and innovation and entrepreneurial ability. For example, the knowledge and skills accumulated in the learning process may provide more theoretical support and practical ability for students to participate in innovation and entrepreneurship projects. Additionally, students who actively participate in innovation and entrepreneurship activities may improve their comprehensive quality in practice, which in turn has a positive impact on their academic performance. Therefore, the positive correlation between GPA and GPI not only reflects the simultaneous growth of students in the two fields, but also may reveal an organic connection between learning and practice in the individual growth of students.

Figure 5 shows the results of the correlation analysis (this figure uses the website: www.chiplot.online). Grade point innovation is divided into six categories: skills examination; arts and sports activities; social practice; arts and sports competitions; disciplinary competitions; and innovation and entrepreneurship practice activities. Since the school has introduced the GPI with a minimum of 6 points, including more than 2 points for student competitions and no more than 2 points for arts and sports, more of the GPI's value can reflect students' performance in competitions. It can also be seen from Figure 5 that disciplinary competition and GPI show a strong positive correlation (0.97). The correlation between sports competition and GPI is weak (0.19). This phenomenon may reflect the fact that schools pay more attention to fields closely related to innovation ability when setting points for innovation and entrepreneurship, and consider arts and sports as supplementary activities that encourage students to develop in a well-rounded way but do not have a significant impact on their GPI. In addition to the negative correlation between Q5 and GPI and GPA, the remaining factors are generally positively correlated or uncorrelated with

GPI and GPA. Of all the factors, Q1 has the most significant correlation, with 0.67 and 0.43, respectively. This shows that the more energy and time students put into participating in innovation and entrepreneurship activities in their spare time, the significantly positive the correlation with the GPI they obtain. The higher the degree of participation in mass entrepreneurship activities, the higher the points they obtain. In addition, the correlation between Q1 and GPA is 0.43; although this correlation is relatively low, it still reflects that reasonable planning and the diversified development of students' extracurricular time may have a certain promoting effect on academic performance.

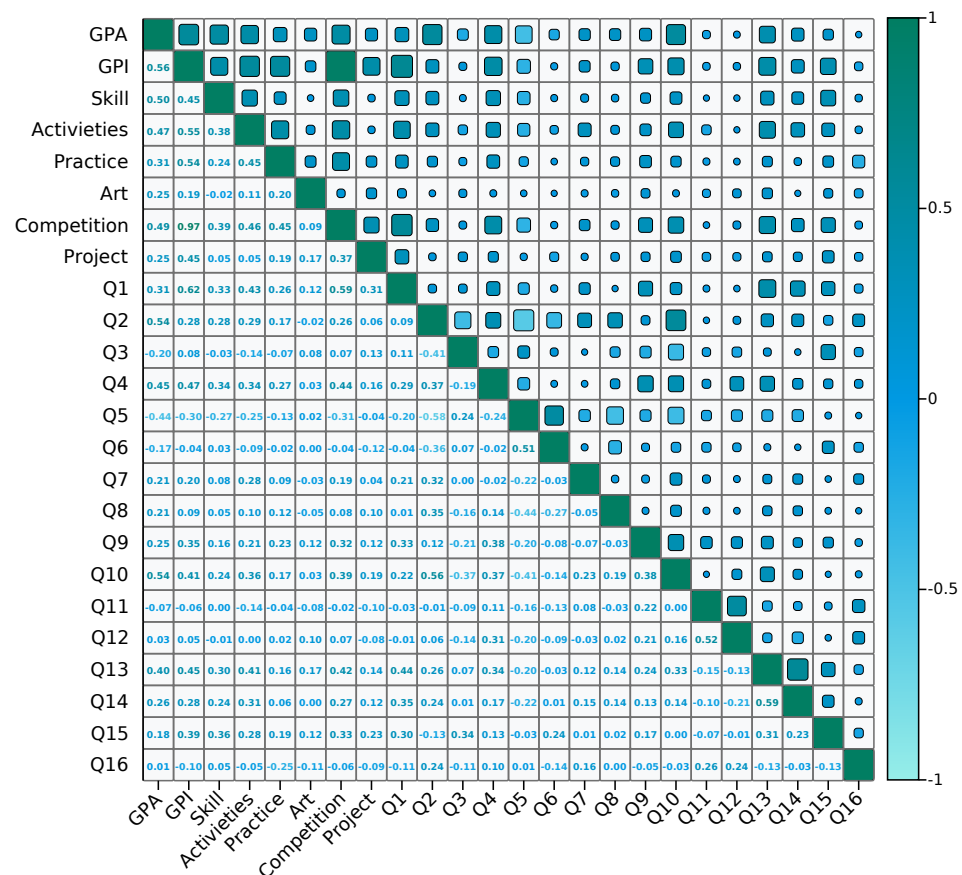


Figure 5. The result of correlation analysis.

3. Modeling of GPI and GPA Based Deep Learning

According to the Pearson correlation coefficient, there are significant correlations between GPI and GPA. Some of the 16 evaluation factors also show a connection with these two key indicators. Therefore, based on these correlations, this section describes how to use the collected valid questionnaire data to build a deep learning predictive model.

3.1. The Structure of Neural Network

The extreme learning machine (ELM) [21] is a kind of single-hidden-layer feed-forward neural network. Its biggest characteristic lies in the extreme nature of its training process. Specifically, ELM does not update the network parameters through the traditional gradient descent algorithm, but randomly initializes the hidden layer weight and bias, directly obtaining the output weight by solving the linear equation, and thus greatly speeding up the training process.

The extreme learning machine autoencoder (ELM-AE) is a new neural network method that can reconstruct input data like an autoencoder (AE). The ELM-AE consists of three layers of networks, namely an input layer, a hidden layer, and an output layer. Its structure

is shown in Figure 6a. The ELM-AE inherited the efficient learning ability of ELM, reconstructing the input data through unsupervised learning, reflecting the core function of AE. Given a dataset containing d samples $x_i (i = 1, 2, \dots, d)$, the hidden layer output of ELM-AE is

$$H = g(w_m x + b_m), w^T w = I, b^T b = I \quad (1)$$

The expression for calculating the output weight is

$$\beta = Y^T (H^T H)^{-1} \quad (2)$$

The deep extreme learning machine (DELM) is a method that combines deep learning and extreme learning machine (ELM) to solve complex machine learning tasks, especially in classification and regression problems of high-dimensional data. The unsupervised learning algorithm ELM-AE is used to initialize the parameters of each layer of the network. The subsequent tedious fine-tuning process is avoided, thus greatly reducing the time and calculation cost required for model training. This unique training mechanism gives DELM a distinct advantage in training efficiency and speed. Compared with other deep learning algorithms, DELM shows excellent performance, not only speeding up the training process of the model but also maintaining the prediction accuracy of the model. The deep kernel extreme learning machine (DKELM) [22] uses kernel function calculation to replace the inner product operation of high-dimensional space and maps the output features of the K th hidden layer through the kernel function, thus achieving feature mapping to the high-dimensional space for decision-making, which is conducive to improving the classification accuracy of DELM. The DKELM structure is shown in Figure 6c.

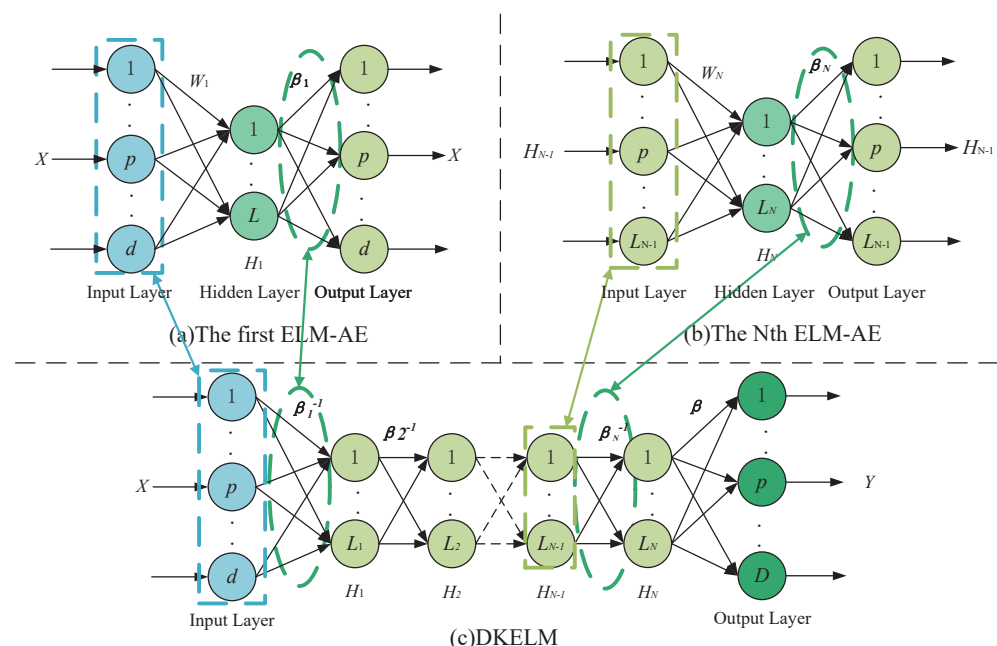


Figure 6. The structure diagram of DKELM.

DKELM first uses ELM-AE to effectively extract the features of the input sample data layer by layer so as to easily distinguish the types with high complexity. After feature extraction in the k hidden layer, the kernel function is used to map the features learned in the previous layer to the high-dimensional space, thus avoiding the calculation of the inner product operation in the high-dimensional space. In this paper, the DKELM method is used to predict the collected questionnaire data. Through this method, the constructed deep neural network model is able to not only improve the processing ability of data but also

effectively improve the prediction accuracy. Especially in the face of high-dimensional and complex types of data, DKELM shows its advantages in feature extraction, nonlinear modeling, and training efficiency, providing strong support for the construction of GPI and GPA prediction models.

3.2. Simulation and Analysis

The scores of 16 evaluation factors are used as input variables. GPA and GPI are set as two output variables to explore the relationship between these factors and academic performance and grade point innovation through deep neural network models. In order to ensure the validity and accuracy of the model, three methods—backpropagation neural network (BP) [23], autoencoder (AE) [24], and support vector machine (SVM) [25]—are chosen for comparison with deep kernel extreme learning machine (DKELM). Among them, the BP neural network is a classical feed-forward neural network that continuously adjusts the network weights through the backpropagation algorithm, enabling the model to minimize prediction error. AE is an unsupervised learning method that is able to capture the potential features of the data by compressing and reconstructing the input data and can be used as a feature extractor to improve prediction ability. SVM is a supervised learning method that is suitable for handling high-dimensional data, capable of maximizing the inter-class spacing by finding the optimal hyperplane, and suitable for regression and classification problems. DKELM is a new type of deep learning method which, by combining the extreme learning machine and kernel method, can effectively improve the learning ability of the model and its ability to deal with complex nonlinear problems, as well as capture the interrelationships between more complex factors so as to achieve accurate prediction of students' academic performance and innovation ability.

In order to comprehensively assess the performance of the model, two evaluation indexes are used to measure the gap between the prediction results and the actual values, namely the coefficient of determination (R^2) and the mean absolute percentage error (MAPE). The R^2 determination coefficient is an evaluation index for linear model evaluation. The closer the value is to 1, the better the model is. The closer the value is to 0, the worse the model. MAPE is the percentage form of error, which can visually show the proportion of the prediction result in relation to the error of the actual value. The lower the value of MAPE, the more accurate the model prediction is. The lower the value of MAPE, the smaller the relative prediction error of the model. The experimental results are shown in Table 1.

Table 1. Prediction results of different deep neural network models.

	BP		AE		SVM		DKELM	
	R^2	MAPE	R^2	MAPE	R^2	MAPE	R^2	MAPE
GPI	0.9821	29.14%	0.6493	37.29%	0.9858	52.19%	0.9937	12.48%
GPA	0.9683	4.488%	0.7134	29.28%	0.9741	24.29%	0.9877	2.659%

From the experimental results, it can be seen that the DKELM method performs well in predicting GPA and GPI, and achieves more excellent prediction results. In addition, by plotting the error box plots between the predicted values and the actual values of all the samples obtained by the four methods, as shown in Figure 7, it can be seen that the DKELM method also has a certain advantage in terms of stability of the prediction results and error. Compared with other methods, DKELM not only performs excellently in terms of accuracy, but also provides more stable and reliable prediction results in practical applications, making it suitable for the scientific assessment and decision-making support of students' academic and innovation abilities.

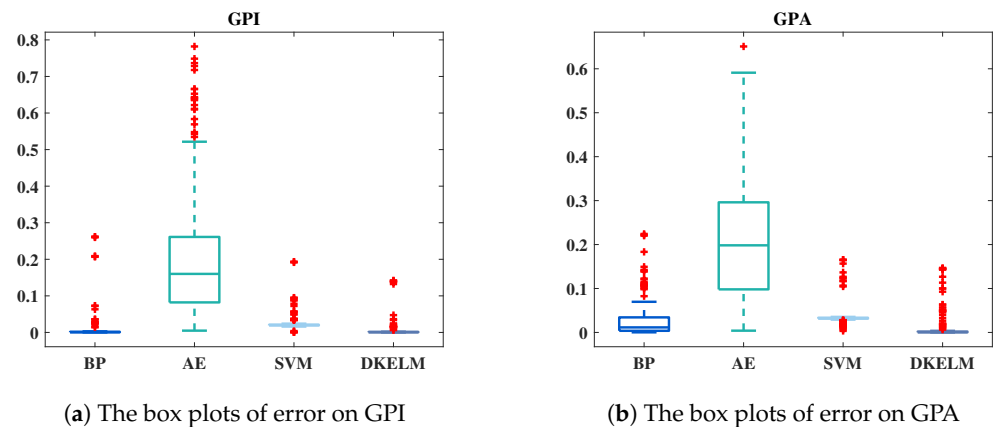


Figure 7. Error box plots for 4 methods.

4. Multi-Objective Decision-Making

In this subsection, GPA and GPI are used as objective functions for multi-objective evolutionary optimization. These two indicators are used to design learning strategies that help students achieve higher grades using an evolutionary algorithm. Students learn a variety of different resources during the learning process, which results in a large number of learner–learning resource interactions where a learner selects multiple learning resources. Therefore, students have a one-to-many relationship with learning resources. The evolution of learning resources in smart learning environments needs to meet the learning needs of learners and help students build a complete knowledge system and knowledge structure.

4.1. The Process of Multi-Objective Decision-Making

In a single-objective optimization problem, there is only one optimization objective, so any two solutions can be directly compared and a unique optimal solution can be determined. Unlike traditional single-objective optimization, multi-objective optimization is carried out when multiple objectives need to be optimized at the same time. Since there may be conflicts between different targets, optimizing one target often leads to performance degradation in other targets, making it difficult to obtain a unique optimal solution [26]. The solution is often a trade-off and a compromise between goals, making the overall goal as optimal as possible. From the correlation analysis, it can be concluded that GPI and GPA are two related factors. Therefore, this study takes GPI and GPA as two objective functions and uses the multi-objective optimization method to solve the learning strategy for improving students' innovation and entrepreneurship abilities.

The 16 evaluation factors are taken as decision variables, while GPI and GPA are taken as two objective functions. The population is first initialized, and then the objective function value corresponding to each population individual is calculated through the constructed DKELM neural network model. The child population is obtained through non-dominated sorting, selection, crossover and mutation, and the parent and child generations are combined as a new population for the next iteration [27]. The optimal solution is the learning strategy corresponding to the higher GPA and GPI values. The multi-objective decision-making process is shown in Figure 8.

Through the above methods, the dynamic evolution of learning resources can be realized according to the learning needs of learners, the scope of resource evolution can be narrowed, and computing time can be saved. The optimal solution obtained by the multi-objective optimization algorithm corresponds to the scores of the 16 evaluation factors, and the weights of different strategies can be obtained according to the results.

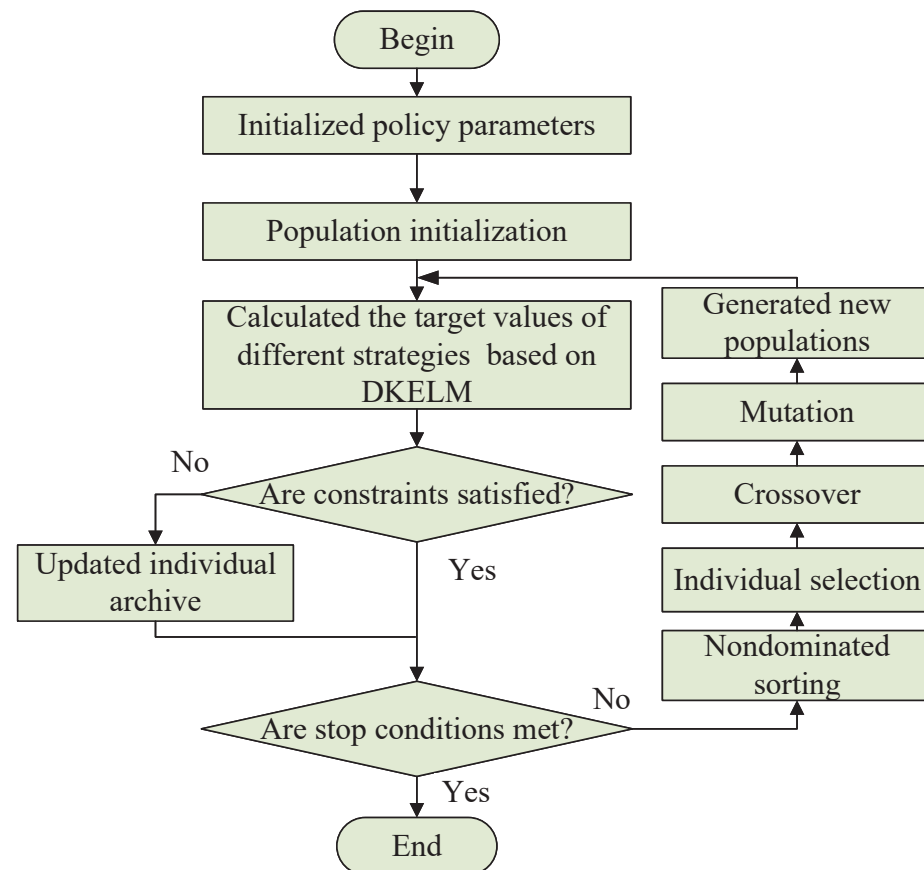


Figure 8. Flow chart of the multi-objective decision-making process.

4.2. Parameter Setting and Experimental Results

In this paper, the algorithm population is set to 100, the number of iterations is set to 1000, and the parameters for crossover and mutation are set as follows: $p_c = 0.9$, $n_c = 20$ (p_c is the crossover probability, n_c is the crossover parameter); and $p_m = 1/D$, $n_m = 20$ (p_m is the mutation probability, n_m is the mutation parameter). Using hypervolume [26] as an evaluation index, the convergence and diversity of the solution set are measured by calculating the hypervolume formed between the solution set and the reference point in the target space. The larger the hypervolume value, the better the result obtained by the algorithm. Figure 9a shows the frontier distribution of the optimal solution obtained by the multi-objective optimization algorithm; Figure 9b shows the calculation results of the hypervolume value in the iterative process of the algorithm. It can be seen that the algorithm used has obtained a good solution set and that the algorithm has good convergence.

After averaging 5% of the optimal value of the strategies solved with different weights, algorithm strategies for different goals are obtained, as shown in Figure 10. In order to obtain higher GPI results, Q4–Q6 strategies should be chosen. To obtain higher GPA results, Q11 and Q12 strategies should be focused on. Although Q16 shows a negative correlation in the correlation matrix, it has no significant difference in terms of its impact on GPA and GPI in the process of designing innovation education strategies.

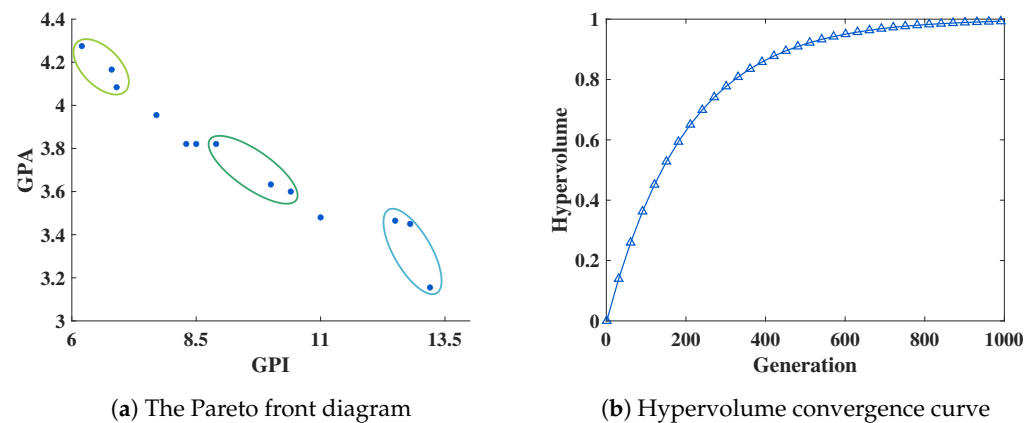


Figure 9. Pareto front diagram and hypervolume convergence curve of the multi-objective optimization algorithm.

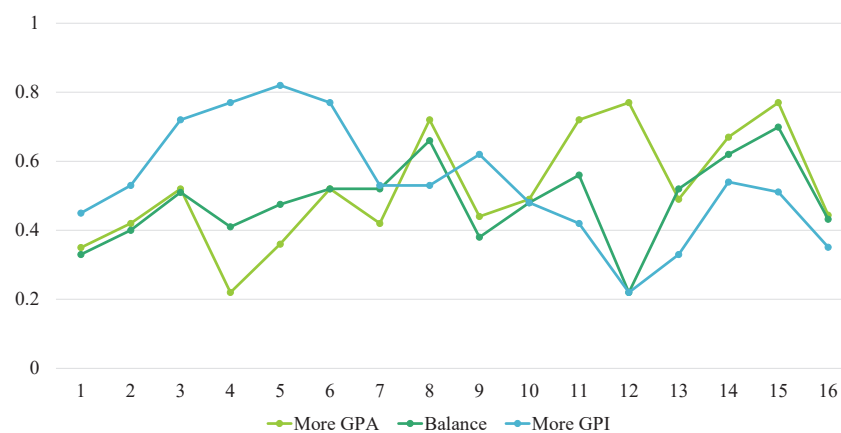


Figure 10. Control strategies for different types.

4.3. Analysis and Discussion

On this basis, we conduct a detailed analysis of students from different grades, majors, and achievement levels. Firstly, regarding grade level, juniors and seniors have accumulated substantial professional knowledge and practical experience due to their relatively stable academic tasks, enabling them to generally meet GPI requirements. They also exhibit high enthusiasm for participating in innovation activities and possess a certain degree of innovation capability. In contrast, freshmen and sophomores are only beginning to engage with university-level professional courses, resulting in insufficient knowledge reserves and significant academic pressure. Consequently, their participation in innovation activities is relatively low, often remaining at the initial stage. To address this situation, it is recommended that schools intensify GPI training, establish targeted innovation practice projects, and provide additional platforms for practice. These measures would assist lower-grade students in gaining early exposure to and understanding of innovation activities, thereby enhancing their innovation awareness and interest in participation, and facilitating their entry into innovation practice more swiftly.

Secondly, from the professional point of view, there are great differences in the participation of students from different disciplinary backgrounds in innovation activities. Due to the strong practicality and technology of engineering majors, students are generally more active in participating in innovation activities, especially in the fields of scientific and technological innovation and engineering practice, where students have more opportunities to conduct research and project development. In contrast, students majoring in liberal arts mainly focus on theoretical study, literature, history, philosophy, and other fields, and have

relatively few opportunities to participate in innovation activities, and the diversity of innovation activities is relatively weak. Therefore, for students of different majors, more innovation activities suitable for the characteristics of liberal arts majors can be designed, such as innovation projects in literary and artistic creation, social research, cultural inheritance, and other fields, to stimulate the innovation potential of liberal arts students.

Finally, from the point of view of achievement level, there is also a certain correlation between students' academic performance and their enthusiasm and ability to participate in innovation activities. Due to the accumulation of a solid foundation of academic knowledge, strong thinking ability, and innovation potential, students with excellent academic performance can usually show a more prominent talent advantage in innovation activities. Students with average or slightly lower grades may have some difficulties in traditional academic courses, but they are often able to make up for their academic shortcomings through hands-on operations, teamwork, and other ways in practical innovation activities, and achieve better results. Therefore, schools should focus on providing personalized innovation education and practice opportunities for students with different performance levels, encourage students with low academic performance to improve their comprehensive ability through innovation activities, and provide higher-level and more challenging innovation platforms for students with excellent performance to promote the all-round development of all students.

Through the comprehensive analysis of students in different grades, majors, and achievement levels, schools can more accurately formulate innovation education policies, maximize the innovation potential of students, and promote the cultivation of innovation talents.

5. Suggestions for University Innovation Education Policy

Based on the analysis and discussion of multi-objective decision-making for different needs in Section 4.3, the following suggestions regarding innovation education are presented:

(1) **Course setup.** The course content and teaching process should not only include traditional professional education and principles of innovation problem-solving but also pay attention to the cultivation of innovative thinking, applications of innovation theory, and practical training in innovation ability. The process of cultivating innovative thinking emphasizes taking common things in life as examples that clarify the law of technological system evolution. For example, in the elective courses of electrical students, the functional model of a technical system is introduced and component cutting is implemented through the case of electric bicycle motor. In practical teaching, teachers can stimulate students' learning interest and motivation by setting learning objectives, putting forward innovation questions, and developing learning environments. Entrepreneurial training sessions can abandon traditional commercial canvas methods and take the mode of college students' innovation and entrepreneurship competition as the model, where students are assigned to teams and topics such as team formation, discussion, writing and defense, etc.

(2) **Teacher ability.** Teachers should first have experience in participating in innovation and entrepreneurship competitions or engaging in entrepreneurial activities. Schools may consider focusing on improving the guidance ability of full-time teachers in innovation and entrepreneurship competitions. Schools may first cultivate an innovative and entrepreneurial atmosphere among teachers by hiring high-level enterprise mentors to guide young teachers' growth, assisting teachers to help improve student projects through venture capital investment, and selecting young teachers to offer innovation and entrepreneurship courses. At the same time, this is considered to give a certain tilt to innovation and entrepreneurship teachers in terms of title evaluation, performance reward, and undertaking

projects. Instructors should fully understand the innovation and entrepreneurship policies at all levels, the requirements of innovation training programs, and the rules of innovation and entrepreneurship competition. The supervisor should also set posts for students to choose topics from their own research direction, and guide students to fully understand innovation and entrepreneurship activities.

(3) **Evaluation of ability.** Innovation and entrepreneurship ability is mainly assessed to improve students' innovative thinking and help them master innovation methods through learning textbook knowledge, overcoming the psychological barrier of 'difficulty in invention and innovation', and stimulating students' enthusiasm for creativity. By studying the problem analysis and problem identification tools of TRIZ, students are able to provide an essential description of the functions of technical systems, master the methods of functional analysis, and create functional model diagrams.

(4) **Innovation practice.** The assessment of innovation and entrepreneurship level should also pay attention to practical ability. The practical activities of innovation and entrepreneurship are not only the starting point of innovation and entrepreneurship education but also the foothold of innovation and entrepreneurship education. Consciousness training and knowledge construction should be assessed based on practical activities. When designing the assessment method of innovation and entrepreneurship, innovation and entrepreneurship competitions and business exercises should be included, and peer evaluation (that is, evaluation by students) should be introduced on the basis of teacher evaluation. Qualified universities can invite outstanding alumni, successful entrepreneurs, and venture capitalists to participate in online blind evaluation and roadshow defense.

(5) **Innovation projects.** Schools should fully encourage students to conduct large-scale project-style training in professional courses and extracurricular activities, and transform innovation projects into "alchemy stones" of innovation and entrepreneurship education. On the one hand, professional courses include course reports, production practice, course design, and other course practice links, and the assessment content of these links may be consistent according to the teaching syllabus for several consecutive years. From the perspective of professional teaching, incorporating new knowledge, new technology, and new ideas into courses can promote best practices that can be applied to college students' innovation and entrepreneurship training programs and other funded projects to improve the overall educational and group content. On the other hand, when students apply for extracurricular programs such as college student innovation and entrepreneurship training programs, they lack topic selection. On the basis of topic selection obtained from full-time tutors, they can supplement topic selection with course practice. By continuously funding high-level student projects and using them as an 'incubator' for student competitions, this approach not only broadens the scope of innovation and entrepreneurship education but also facilitates the selection of high-level teams to participate in competitions.

(6) **Interdisciplinary communication.** Student teams should not only maintain complementary talent in each position, but should also have the ability to consistently attract the best talent to join them. Through the division of the establishment of posts, the same team maintains the inter-generational inheritance among team members. This gives full play to the advantages of senior students in terms of professional ability and business level and ensures that the student team can recruit enough excellent students. Innovation and entrepreneurship departments and grassroots teaching units can invite outstanding student teams to enter innovation and entrepreneurship studios according to their level and provide technical guidance and practice venues. While the teams make their own blood, the student work department and the innovation and entrepreneurship department can jointly organize the participation of junior students in the division of labor and set up posts to supply blood for the student teams.

6. Conclusions and Prospect

Conclusion. First of all, this study emphasizes that teachers should have experience in innovation and entrepreneurship and cultivate their guiding ability through various means. At the same time, teachers are encouraged to have an in-depth understanding of innovation and entrepreneurship policies and guide students to participate in innovation and entrepreneurship activities. Secondly, the curriculum focuses on the combination of traditional professional education and innovation thinking training, stimulating students' creative enthusiasm through practical teaching and innovation and entrepreneurship competition modes. In terms of ability evaluation, the main assessment of students' innovation and entrepreneurial ability should pay attention to the investigation of practical ability and introduce diversified evaluation methods. In addition, students are encouraged to conduct large-scale project-based training to transform innovation projects into results of innovation and entrepreneurship education. Finally, interdisciplinary education is regarded as an important way to improve the effect of innovation and entrepreneurship education and to promote the innovation development of student teams by optimizing team structure and providing technical support.

Limitations. There are still some issues in the design of innovation and entrepreneurship education. Innovation and entrepreneurship education and professional education are disconnected to varying degrees, and professional teachers teach students around professional courses, mainly guiding graduate students to engage in scientific research, and further attention needs to be paid to the innovation practice activities of undergraduates. Some college teachers have a high level of scientific research ability but do not know enough about college students' innovation and entrepreneurship training programs and competitions, thus failing to inspire an effective reaction with students. In universities, grassroots colleges and innovation and entrepreneurship guidance departments, university science parks, and technology transfer centers need to be more closely connected. Poor communication makes it difficult to translate students' innovation consciousness into practical innovation abilities.

Future research. In the future, how to effectively open up all aspects of innovation and entrepreneurship education through the formulation of an evaluation system will be an important research topic. The evaluation system of innovation and entrepreneurship education will be more diversified and comprehensive, focusing not only on students' innovation and entrepreneurship results but also on the evaluation of their innovation thinking, teamwork, problem-solving, and other abilities in the process. At the same time, more external experts and enterprises will be introduced to participate in the evaluation to improve the objectivity and practicality of the evaluation.

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Abbreviations

The following abbreviations are used in this manuscript:

GPA	Grade point average
GPI	Grade point innovation
ELM	Extreme learning machine
ELM-AE	Extreme learning machine autoencoder
DELM	Deep extreme learning machine
DKELM	Deep kernel extreme learning machine
BP	Backpropagation neural network
AE	Autoencoder
SVM	Support vector machine
MAPE	Mean absolute percentage error

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